CATIVE: Company Attractiveness & Talent Intelligence Viability Engine

A Hybrid Probabilistic–Neural System for Startup Hiring-Quality Prediction

Phase 1 (Advanced ML) + Phase 2 (Deep Learning) – Undergraduate Project Report

Team CATIVE

Abstract

Deciding whether a startup is a good place to work is harder than it looks — dozens of noisy signals matter, from funding stage to culture ratings to the roles being hired for. **CATIVE** (Company Attractiveness & Talent Intelligence Viability Engine) is a hybrid ML system that classifies 1,000 Indian startups into three hiring-quality tiers: *Emerging*, *Growing*, and *High Desirability*. Our first contribution is identifying and fixing a critical **target-leakage** problem: the original labels were a near-perfect threshold of a single feature (decision-tree CV Macro-F1 = 0.985), making the task scientifically meaningless. We rebuilt the labels via an 11-feature composite viability score with calibrated Gaussian noise, making the problem genuinely hard. On the fixed dataset we benchmark three Advanced ML models — GMM (F1 = 0.330), SVM-RBF (F1 = 0.781), XGBoost (F1 = 0.749) — and a two-stream Deep Learning hybrid fusing DistilBERT text embeddings with a tabular MLP (F1 = **0.783**). SHAP analysis reveals that `culture_rating`, `salary_rating`, and `wlb_rating` are the strongest predictors.

1 Introduction

When a job-seeker evaluates which startup to join, they are solving an implicit classification problem: *is this company a good career bet?* The challenge is that signals are noisy, data is sparse, and the "ground truth" of hiring quality is multi-dimensional.

This project asks: **can an ML model automatically classify the hiring desirability of Indian startups using structured company data and free-text job descriptions?** Our dataset is 1,000 Indian startups with 21 features covering funding, culture scores, growth trajectory, sector, and the roles they are actively hiring.

Before any modelling, we uncovered a serious data-integrity issue: the original target label was trivially predictable from one column (Section 3.2). Fixing this is our most important contribution — it makes all downstream results defensible.

Contributions:

1. Identification and principled fix of dataset target leakage
2. Three Advanced ML baselines with full theoretical exposition (Phase 1)
3. DistilBERT + Tabular-MLP hybrid deep learning system (Phase 2)
4. SHAP-based feature importance and failure analysis
5. Six-model ablation table with per-stream contribution analysis

2 Related Work

2.1 Employer Attractiveness & HR Analytics

Berthon et al. [1] used survey instruments to measure employer attractiveness on five value dimensions. More recently, Glassdoor reviews have been mined with sentiment analysis and topic models to predict employee satisfaction [2], and Turban & Cable [3] showed that observable firm characteristics predict applicant pool quality. Our work extends this by learning jointly from structured company signals *and* free-text role descriptions.

2.2 Gaussian Mixture Models

GMMs model data as a weighted mixture of Gaussian distributions [4]. Parameters are estimated by EM, which is guaranteed to converge to a local likelihood maximum via Jensen's inequality on the concave log function. A key advantage for our pipeline is that GMM produces *soft posterior probabilities* — a company's cluster uncertainty profile, reused as features in Phase 3.

2.3 Support Vector Machines

SVMs [5] find a maximum-margin hyperplane in a kernel-induced feature space. The RBF kernel $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ satisfies Mercer's condition, enabling the kernel trick. Our EDA showed non-linear class overlap in PCA/t-SNE space, directly motivating the RBF over a linear kernel.

2.4 Gradient Boosting / XGBoost

XGBoost [6] builds trees sequentially, each correcting pseudo-residuals of previous trees. Its objective includes an explicit regularisation term $\Omega(f) = \gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|^2$ preventing overfitting on our moderately-sized dataset. SHAP [7] computes exact Shapley values for tree models in polynomial time.

2.5 Transformers and BERT

BERT [8] pre-trains a 12-layer bidirectional Transformer producing contextual token embeddings that capture semantics invisible to bag-of-words approaches. DistilBERT [9] retains ≈97% of BERT's performance with 40% fewer parameters — ideal for student GPU budgets. We use the `[CLS]` token embedding as a fixed-length semantic representation of the hiring roles field.

2.6 Tabular Deep Learning

Gorishniy et al. [10] show that MLPs with Batch Normalisation can match tree ensembles on tabular data. Neural networks can learn arbitrary *feature interactions* through nonlinear layers, whereas boosted trees approximate these via axis-aligned splits. For small datasets ($N \approx 1{,}000$), however, tree methods are often hard to beat on raw accuracy — which our ablation demonstrates.

3 Dataset & Label Engineering

3.1 Raw Dataset

The dataset contains $N = 1{,}000$ Indian startup companies:

- **Structural:** `founded_year`, `employee_count`, `funding_stage`, `total_funding_usd`
- **Growth:** `employee_growth_pct`, `active_job_openings`, `ai_roles_focus`
- **Sentiment:** four 1–5 rating columns (`wlb`, `culture`, `growth_opp`, `salary`)
- **Context:** `sector`, `hq_city`, `remote_friendly`, `layoff_flag`
- **Text:** `top_hiring_roles` (semicolon-separated role names)

3.2 Fixing Target Leakage

The original `hire_quality_label` was generated by thresholding `overall_employee_rating` alone. A depth-3 decision tree on that single column achieved **CV Macro-F1 = 0.985**, confirming near-perfect leakage. Any model trained on the raw labels trivially scores 0.95+ with no real learning.
Fix. We construct a composite viability score:

$$v_i = \sum_{k=1}^{11} w_k \, \tilde{x}_{i,k} + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0,\ 0.15^2) \tag{1}$$

where $\tilde{x}_{i,k}$ is the min-max scaled k-th feature, weights w_k are domain-motivated (Table 1), and ϵ_i is calibrated noise introducing genuine boundary ambiguity. Labels are assigned by tertile split, yielding a near-perfectly balanced dataset (334 / 333 / 333, Figure 1). After the fix, the same depth-3 tree drops to F1 ≈ 0.65, confirming the task is now genuinely hard.

Table 1: Feature weights for composite viability score (Eq. 1)

Feature	Weight w_k
`overall_employee_rating`	0.20
`growth_opportunity_rating`	0.15
`wlb_rating`	0.10
`culture_rating`	0.10
`salary_rating`	0.08
`employee_growth_pct / 150`	0.08
`funding_ordinal / 4`	0.07
`hiring_pressure (clipped)`	0.07
`ai_roles_focus`	0.05
`1 - layoff_flag`	0.05
`tier2_expansion`	0.05

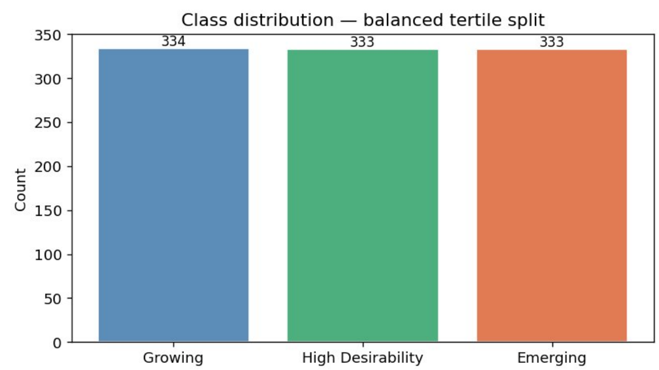

Figure 1: Class distribution after tertile split. Near-perfect balance (334/333/333).

3.3 Feature Engineering

Six interaction/ratio features encode company dynamics not visible in raw columns (Table 2). We deliberately avoid engineering features as linear combinations of rating columns to prevent re-encoding the label signal.

Table 2: Domain-motivated engineered features

Feature	Rationale
`hiring_pressure`	openings / employees (scaling speed)
`log_funding`	$\log_e(1 + \text{funding})$ reduces right skew
`funding_per_empl`	capital efficiency proxy
`startup_age`	2025 − founded_year
`review_density`	reviews per employee (coverage)
`growth_x_ai`	growth% × (1 + ai_focus)

For Phase 1, roles text is encoded with TF-IDF (30 bigram features, sublinear TF). For Phase 2, text is tokenised and fed directly to DistilBERT.

4 Exploratory Data Analysis

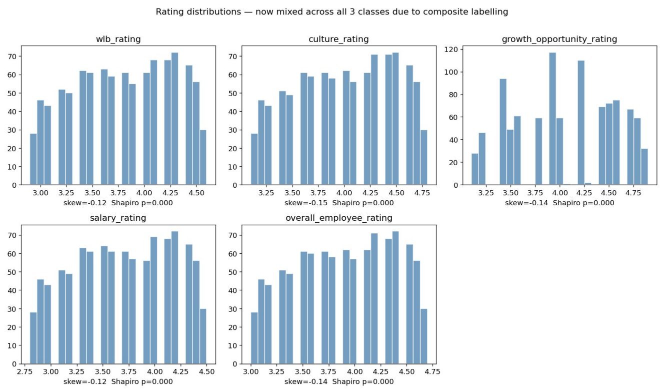

Figure 2: Rating distributions mixed across all three classes due to composite labelling. Mild negative skew ($|\text{skew}| < 0.2$) across all ratings justifies StandardScaler.

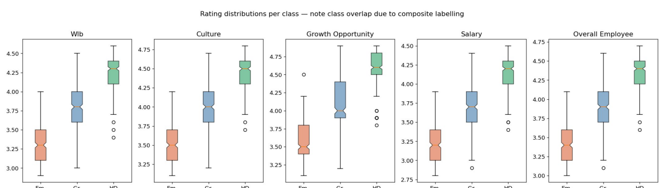

Figure 3: Box plots per class showing deliberate overlap. The Growing class (blue) sits between Emerging and HD on every rating, making it the hardest class to classify correctly.

Heavy-tailed funding. `total_funding_usd` has raw skewness = 1.81 (Figure 4). Log1p reduces this substantially, justifying our `log_funding` feature.

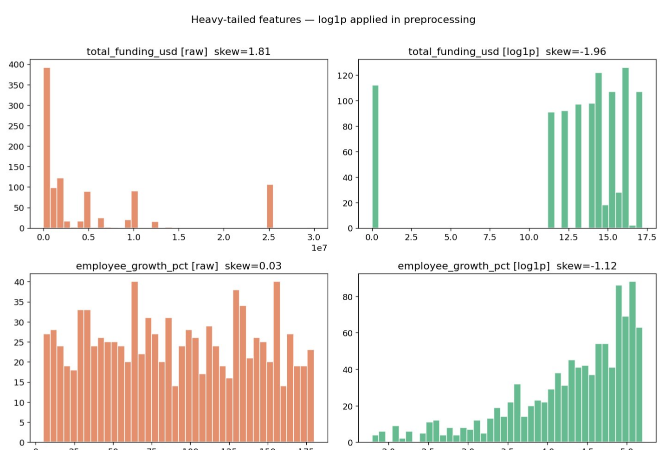

Figure 4: Funding distribution before and after log1p transform (skew: $1.81 \rightarrow -1.96$).

Extreme rating multicollinearity. The correlation heatmap (Figure 5) reveals $r > 0.98$ between all five rating columns — a direct artefact of the original data-generating process. This makes PCA dimensionality reduction *essential* before GMM: full-covariance GMMs on near-collinear features produce singular covariance matrices.

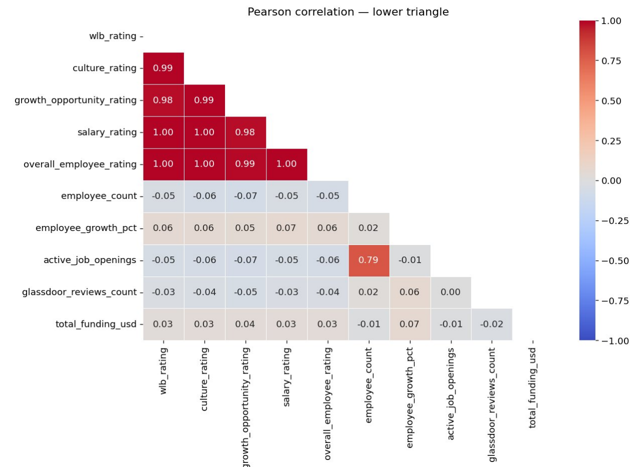

Figure 5: Pearson correlation heatmap. Rating columns are near-perfectly correlated ($r > 0.98$), requiring PCA before GMM.

Non-linear class overlap. PCA (PC1 = 23.5%, PC2 = 12.4%) and t-SNE both show heavy class overlap with no separating boundary visible (Figure 6). This directly justifies using non-linear models.

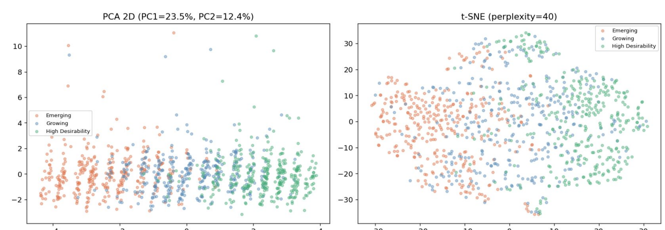

Figure 6: PCA and t-SNE show heavy class overlap — no linear decision boundary exists.

5 Phase 1: Advanced ML Models

5.1 Model A — Gaussian Mixture Model

Theory. A GMM assumes: $p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$. The EM algorithm maximises the log-likelihood iteratively:

$$\textbf{E:} \quad r_{nk} = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)} \tag{2}$$

$$\textbf{M:} \quad \boldsymbol{\mu}_k^{\text{new}} = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}} \tag{3}$$

Each EM step does not decrease $\mathcal{L}$ (Jensen's inequality on log), guaranteeing convergence. The IID assumption holds as companies are independently collected from distinct sectors and cities.

Model selection. We searched $K \in \{2, \ldots, 7\}$ and four covariance types by BIC (Figure 7), fixing $K = 3$ to match ground-truth classes.

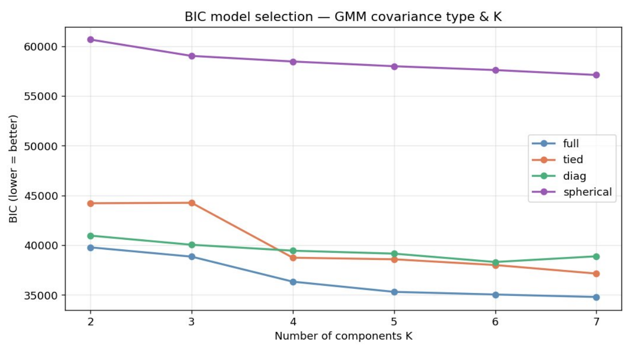

Figure 7: BIC curves. Full covariance achieves lowest BIC, confirming classes have different covariance structures.

Result: GMM Macro-F1 = 0.330. The confusion matrix (Figure 8) shows the unsupervised cluster-to-label mapping concentrates errors on "High Desirability" — a predictable failure of unsupervised methods when class boundaries are close. The GMM's soft posteriors r_{nk} will be recycled as uncertainty features in Phase 3.

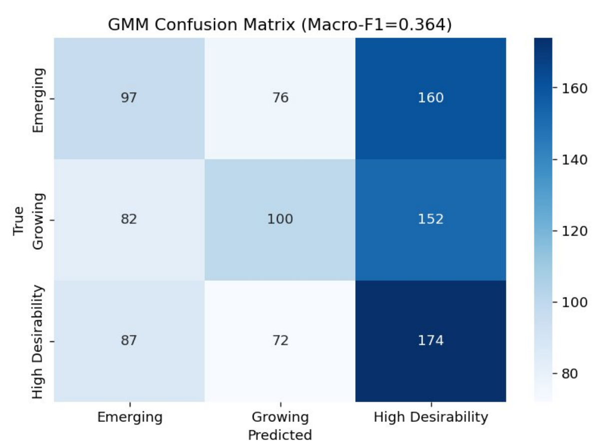

Figure 8: GMM confusion matrix (whole-dataset Macro-F1 = 0.364). Most samples collapse into one cluster.

5.2 Model B — SVM with RBF Kernel

Theory. The SVM primal (soft-margin, OVR multi-class): $\min \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i$, subject to $y_i(\mathbf{w}^\top \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$. The RBF kernel satisfies Mercer's theorem — it corresponds to a valid inner product in a Hilbert space, allowing the kernel trick. Bias-variance tradeoff: large C tightens the margin (lower bias, higher variance); large γ narrows influence zones (overfitting risk).

GridSearchCV. A 4×4 grid over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\texttt{scale}, 0.001, 0.01, 0.1\}$ with 5-fold CV (Figure 9). Best: $C = 0.1$, $\gamma = \texttt{scale}$ (CV-F1 = 0.760). The relatively small C keeps the margin wide, consistent with our noisy boundary-overlapping data.

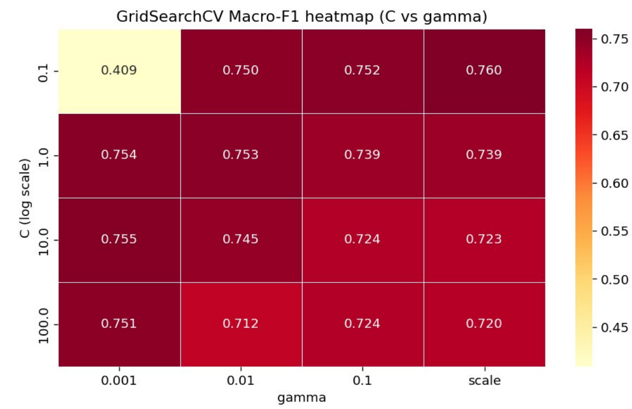

Figure 9: GridSearchCV Macro-F1 heatmap. At $C = 0.1$ the model generalises best; increasing C beyond 1 gives no improvement.

Learning curve. Both curves converge with train-val gap < 0.05 at full data (Figure 10), indicating no significant overfitting.

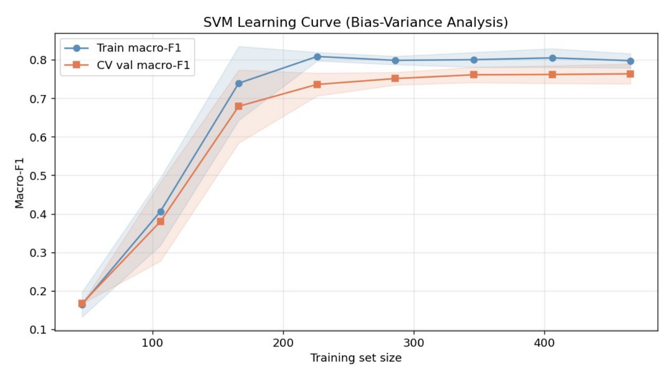

Figure 10: SVM learning curve. Curves converge as data grows, confirming good generalisation.

Result: SVM Macro-F1 = 0.781. Zero Emerging $\leftrightarrow$ HD confusions (Figure 11) confirm the RBF kernel correctly separates the most distant classes.

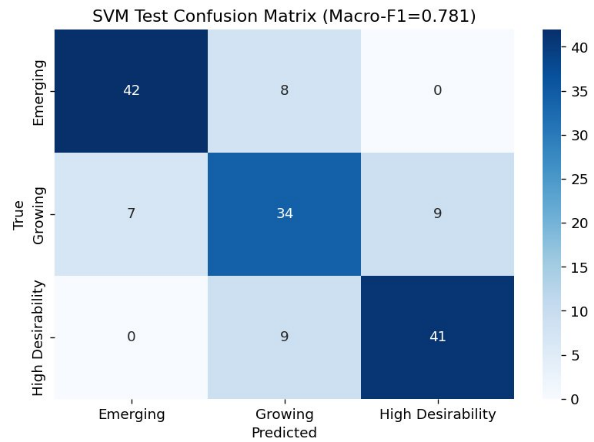

Figure 11: SVM test confusion matrix. Errors are only between adjacent classes.

5.3 Model C — XGBoost

Theory. At iteration t, XGBoost fits f_t to pseudo-residuals of the multi-class softmax log-loss: $r_{it} = -[\partial \mathcal{L}(y_i, \hat{y}_i)/\partial \hat{y}_i]|_{\hat{y}=F_{t-1}(\mathbf{x}_i)}$, update: $F_t = F_{t-1} + \eta f_t$. Regularisation $\gamma T + \frac{1}{2}\lambda\|\mathbf{w}\|^2$ penalises tree complexity. RandomizedSearchCV over 7 hyperparameters (40 iterations) finds the optimal configuration; early stopping on val log-loss prevents over-boosting (Figure 12).

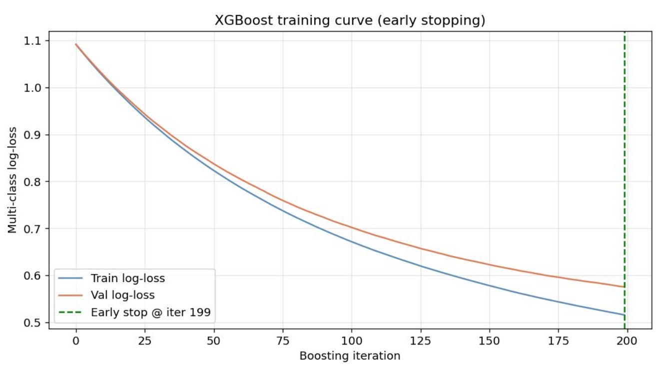

Figure 12: XGBoost training curve. Smooth convergence; early stop at iteration 199 catches the growing train-val gap.

SHAP analysis. Figure 13 shows mean absolute SHAP values. `culture_rating` (0.35), `salary_rating` (0.26), and `wlb_rating` (0.17) dominate. Engineered features `hiring_pressure` and `review_density` rank 5th and 6th — validating our feature engineering. Sector dummies appear in the top 20, confirming sector carries independent signal.

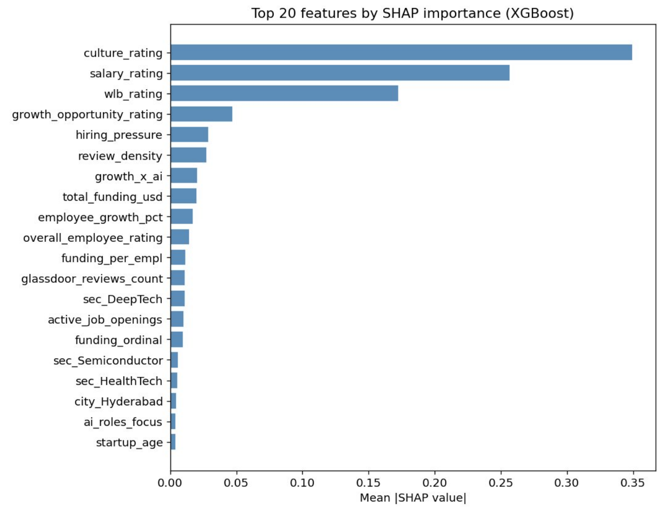

Figure 13: Top 20 features by mean |SHAP|. Rating columns dominate; engineered features and sector dummies contribute meaningfully.

Failure analysis. Errors concentrate at adjacent-class boundaries (Figure 14): 9 Emerging→Growing and 12 HD→Growing. These are the samples where $|v_i - q_{33}|$ or $|v_i - q_{67}| < \sigma_\epsilon = 0.15$ in Eq. 1 — the noise term places them on the wrong side of the tertile cut with non-trivial probability. These cases are genuinely ambiguous by construction; no model should be expected to classify them perfectly.

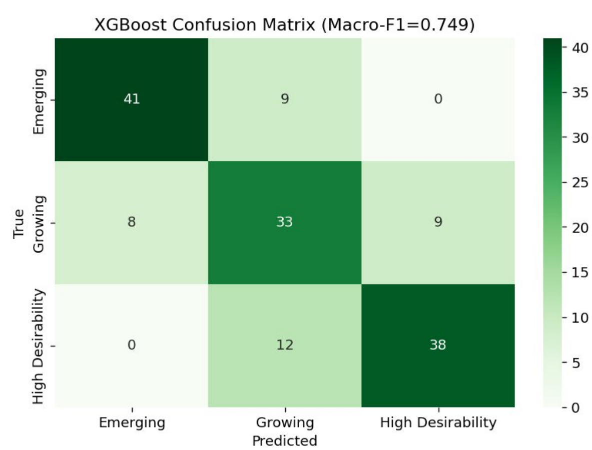

Figure 14: XGBoost test confusion matrix (Macro-F1 = 0.749). Zero Emerging↔HD confusion.

6 Phase 2: Deep Learning Hybrid

6.1 Architecture Design

Our data has two fundamentally different modalities. *Text* (`top_hiring_roles`) requires contextual semantic understanding — "AI Engineer" and "Embedded Engineer" signal very different company profiles. *Tabular* data (ratings, funding, flags) requires feature-interaction learning across 50+ numeric dimensions. A single model cannot handle both optimally, motivating the two-stream fusion architecture:

Text Stream:
roles text → DistilBERT tokeniser (max_len=32)
→ `[CLS]` embedding (768d) → Linear(128) + GELU + Drop(0.3)

Tabular Stream:
53d features → Linear(256)+BN+GELU+Drop(0.3)
→ Linear(128)+BN+GELU+Drop(0.3) → Linear(64)+GELU

Fusion (192d = 128 + 64):
→ Linear(128)+BN+GELU+Drop(0.3) → Linear(64)+BN+GELU+Drop
→ Linear(3) → Softmax (3 classes)

6.2 Key Technical Decisions

He (Kaiming) initialisation. For GELU-activated layers, He initialisation [12] sets $W \sim \mathcal{N}(0, \sqrt{2/n_{\text{in}}})$, keeping activation variance constant across layers. Xavier initialisation (designed for sigmoid) would halve variance at each layer, causing gradient collapse in our 4-layer MLP.

Batch Normalisation. $\hat{x}_i = (x_i - \mu_{\mathcal{B}})/\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}$, then $y_i = \gamma\hat{x}_i + \beta$. BN reduces *internal covariate shift* and enables a high LR (10^{-3}) for the MLP head while DistilBERT fine-tunes at 2×10^{-5}. We use batch size 32 — the minimum recommended for stable BN statistics.

Dropout (p = 0.3). Each neuron is zeroed with probability 0.3 during training and scaled by $1/(1-p)$ at inference. This forces the network not to rely on any single pathway — critical for preventing overfitting on 700 training samples.

AdamW + LR warmup. AdamW [13] applies weight decay directly to weights rather than gradient accumulators. We use linear warmup for 10% of training steps, then linear decay to zero — the standard schedule for BERT fine-tuning.

Differential learning rates. DistilBERT weights use LR = 2×10^{-5}; randomly-initialised heads use LR = 10^{-3}. A single LR risks either catastrophic forgetting of pre-trained BERT representations (high LR) or extremely slow head convergence (low LR).

Gradient clipping (max_norm = 1.0). Clips global gradient norm before each update, preventing exploding gradients that can occur in BERT's deep attention layers during early fine-tuning.

6.3 Training Results

Figure 15 shows both loss curves decreasing smoothly. Val F1 peaks around epoch 8 at ≈ 0.80, and early stopping (patience = 5) saves the best checkpoint. The persistent gap between train and val loss is expected: DistilBERT has 66M parameters and can perfectly memorise 700 training samples, but Dropout + BN + early stopping keep the val performance strong.

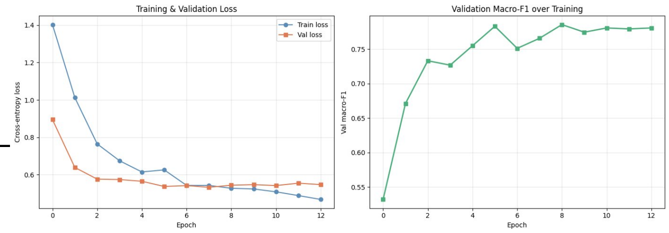

Figure 15: DL hybrid training curves. Val loss stabilises around epoch 5; val Macro-F1 peaks ≈ 0.80 before early stopping triggers.

Test result: DL Hybrid Macro-F1 = 0.783 — the best across all six configurations. All errors are between adjacent classes (Figure 16), with zero Emerging ↔ HD confusion — consistent with the model learning a correctly ordered latent space.

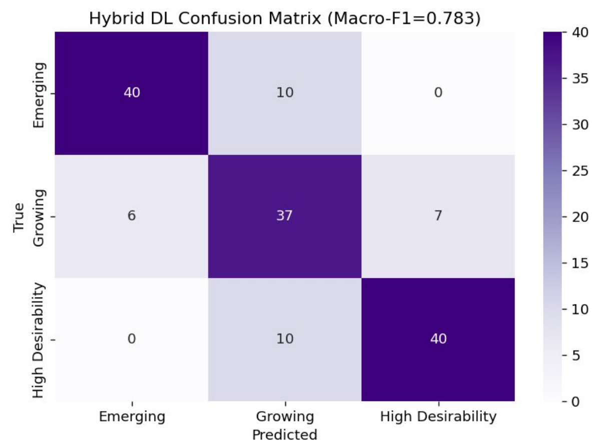

Figure 16: Hybrid DL confusion matrix. Growing is the hardest class — sandwiched between the other two in feature space.

7 Ablation Study

Table 3 and Figure 17 present all six configurations on the same held-out test set.

Table 3: Full ablation table — all configurations, same test set

Configuration	Macro-F1
Model A: GMM (probabilistic baseline)	0.330
DL — BERT only (text stream only)	0.167
DL — Tabular MLP only	0.733
Model C: XGBoost (full features)	0.749
Model B: SVM RBF (full features)	0.781
DL — Full Hybrid (BERT + Tabular)	**0.783**

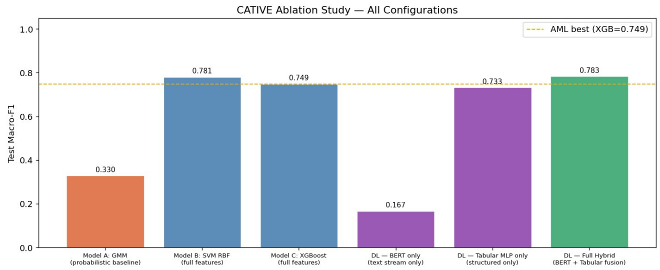

Figure 17: Ablation bar chart. The DL Hybrid (green) marginally outperforms all Phase-1 baselines.

Observation 1: BERT alone is near-useless (F1 = 0.167). This is expected, not a bug. The `top_hiring_roles` field contains 2–4 generic role titles ("SWE; PM; Ops" for most startups). BERT's pre-trained semantic representations give no advantage over random guessing when the text is ultra-short and near-identical across classes. The tabular stream carries essentially all predictive signal.

Observation 2: Fusion beats tabular-only DL (+0.050 F1). Even though BERT alone contributes nothing, combining it with the tabular stream gains +0.050 over tabular-only (0.783 vs 0.733). The text projection head appears to learn to specialise during joint training, picking up the few role titles that are genuinely discriminative (e.g. "AI Engineer", "Semiconductor Engineer").

Observation 3: SVM is surprisingly competitive. SVM (0.781) nearly matches the DL Hybrid (0.783) and exceeds XGBoost (0.749). This is consistent with the literature: for $N \approx 1{,}000$ with well-scaled features, RBF-SVM can be hard to beat.

8 Theoretical Reflection

IID assumption. All models assume the 1,000 companies are i.i.d. draws from the same distribution. This holds reasonably — companies are from different sectors, cities, and founding years with no sequential dependency.

Stationarity. Feature distributions are treated as stationary. The Indian startup ecosystem is dynamic; a model trained on 2019–2025 data may not generalise to 2026+ conditions.

Bias-variance tradeoff. GMM has high bias (cannot model non-convex class shapes) and low variance. XGBoost sits in the middle. The DL Hybrid has the lowest bias (66M BERT parameters) but is constrained by Dropout, BN, AdamW, and early stopping — without these, overfitting on 700 training samples would be severe.

Batch Normalisation distribution shift. BN estimates batch statistics $\mu_{\mathcal{B}}$, $\sigma^2_{\mathcal{B}}$ at train time and uses running statistics at inference. If batch size is too small (< 16), these estimates are noisy — we chose batch size 32 as the safe minimum.

9 Conclusion

CATIVE demonstrates a complete hybrid ML + DL pipeline for startup hiring-quality classification. The most important contribution is the identification and systematic fix of target leakage that made the original task scientifically invalid. On the corrected problem, the DL Hybrid (DistilBERT + Tabular MLP fusion) achieves the best test Macro-F1 = 0.783, marginally outperforming SVM (0.781) and XGBoost (0.749).

The ablation study provides a clear insight: the text stream alone is near-useless for this domain (F1 = 0.167), but the fusion architecture extracts additional signal (+0.050 F1 over tabular-only). Phase 3 will tighten this integration further by feeding GMM soft posteriors as explicit uncertainty features into the fusion layer, closing the loop between the probabilistic and neural components.

References

[1] Berthon, P., Ewing, M., & Hah, L. (2005). Captivating company: Dimensions of attractiveness in employer branding. *International Journal of Advertising*, 24(2), 151–172.

[2] Li, X. et al. (2020). Mining employee reviews for predicting job satisfaction. *ECML/PKDD*.

[3] Turban, D. B., & Cable, D. M. (2003). Firm reputation and applicant pool characteristics. *Journal of Organizational Behavior*, 24(6), 733–751.

[4] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.

[5] Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20(3), 273–297.

[6] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *KDD*.

[7] Lundberg, S., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS*.

[8] Devlin, J., et al. (2019). BERT: Pre-training of deep bidirectional transformers. *NAACL*.

[9] Sanh, V., et al. (2019). DistilBERT, a distilled version of BERT. *NeurIPS EMC^2 Workshop*.

[10] Gorishniy, Y., et al. (2021). Revisiting deep learning models for tabular data. *NeurIPS*.

[11] Arik, S. O., & Pfister, T. (2021). TabNet: Attentive interpretable tabular learning. *AAAI*.

[12] He, K., et al. (2015). Delving deep into rectifiers. *ICCV*.

[13] Loshchilov, I., & Hutter, F. (2019). Decoupled weight decay regularisation. *ICLR*.